



CLIENT NETWORK DESIGN

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Logical and Physical Network Diagrams

The logical and physical diagrams are attached at the bottom of the document

Network Hierarchy Justification

This network design uses a hierarchy instead of a flat design because FR1 requires the network to use a network hierarchy model. A singular device would not be able to handle all the networks functions at an enterprise level. This network needs different things for each zone. For example, ATC needs fixed 10G throughput with NIC failover at the hardware level, where as gate wallboards need only basic 1G throughput. Point of sales need only simple lower bandwidth connection. By implementing network hierarchy, each zone receives what it needs, rather than everything having the same capability. Hierarchy is the mechanism that is responsible for managing and applying redundancy smarter rather than doubling everything to eliminate single points of failure.

For a three tiered model to be justified over a two tiered or collapse core network, there must be three specific criteria met. These criteria include the limitation of not being able to attach more access switches to a singular distribution switch directly, the site of the network is required to reach a vast geographical area, or the distribution to core uplink bandwidth is bottlenecked and congested. From research, it is confirmed that terminal 1 is approximately 800 metres long. This means that the airport requires eight access zones which are separate from each other. These are unable to be aggregated into a single collapsed distribution and core point. The office building on level one is the second reason why a three tier model would be justifiable. NFR2 states that the air traffic control workstations all need simultaneous 10G throughput, thus requiring that part of the network to have its own distribution tier to manage traffic and separate it from other general traffic like from the terminal. A collapsed core would go against this requirement. That is why the most logical application for the network is 4 separate distribution blocks that cover air traffic control, the security cameras, the server farm and the general terminal. A collapsed core would make it impossible to split traffic by type and reduce redundancy.

VLAN Segmentation and Gateway Placement Justification

This network implements VLAN segmentation because FR2 requires for the network to have VLAN segmentation and inter-VLAN routing. With this comes better broadcast domain control, which grants smaller broadcast domains, better network performance

and improved security. There are many different device types on the network, so a single flat broadcast domain would be fragile and inefficient. Segmentation also provides better security separation. Due to the security requirements for network traffic from zones like air traffic control, it is important to make that traffic unreachable from lower priority, public-facing traffic like gate-wallboards and point of sales. The different traffic types also mean that they would need different treatments. For example, CCTV has constant 100Mbps per camera streaming which is listed in NFR5. Whereas air traffic control requires guaranteed 10G, listed in NFR2. For both policies to be implemented, they must be on separate networks, which is only possible via VLAN's. In this network, the gateways will be located at the distribution layer instead of the core. This is because the network should have the distribution switches act as the default gateway for all end users and is generally best practice. This helps free up the load on the core layer devices and ensures that their sole job is layer 3 transit between the WAN edge and distribution blocks. This gives each distribution block its own zone to serve, each with their own switched virtual interfaces so that each zone can have their own redundancy. This makes the network distinct from a router on a stick which would be more suitable for networks on a smaller scale. Because each distribution block has allocated SVIs, this means that each distribution switch will handle its own inter-VLAN routing for their dedicated block.

Distribution Block Redundancy Justification

Instead of a single distribution layer, this network has 4 separate distribution blocks. These areas include air traffic control, the server farm, security cameras and the general area which includes the main terminal. The reason the network has 4 separate distribution blocks is because each of these areas need to be treated differently. Air traffic control is a critical area that requires security and safety and has its throughput and NIC failover requirements listed in NFR2 and NFR6 respectively. The server farm is on its own distribution block because it is a critical service link which relates to NFR7. CCTV will have high amounts of footage streamed through its cameras and grouping them together makes it easier to meet NFR5. The rest of the terminal and other general traffic will be on its own distribution as well. Since the blocks are separated, their individual redundancy can be set. Going into more detail with air traffic control, this distribution block will operate with two switches, this is to meet NFR6 which needs failover redundancy for each workstations network interface card so each workstation will have dual NICs that connect to each switch. Because of this there will be a main ATC connection and a backup ATC connection, each with their own VLAN and gateway. The workstations will decide which of the two NIC's will carry traffic on their independent paths. The workstations also connect to the distribution switches directly

to reduce the amount of hops traffic from the workstations must take. The capacity of these switches are sufficient and will be two 48 port mGig switches for 30 workstations. For the server farm the network is set up to consider NFR7, meeting this requirement helps guarantee that NFR2 is met. Since ATC still relies on the servers, they could be considered a single point of failure. This is why the servers have their own distribution block. All the servers share the same VLAN and both switches are on one subnet. The server block will use an inter-switch trunk between switches and will use hot standby router protocol to select one as active and the other as standby so if a switch fails the switches own servers will go offline but HSRP keeps the gateway up for the other servers. I chose to use this approach because even though this block has two switches like ATC this block does not need two independent paths. The requirement is for the server farm to stay reachable. With this setup, a single VLAN and HSRP allows for one subnet. For the security cameras this network has a singular distribution switch, NFR5 only requires that the network can handle 100 Mbps per camera to the server farm and did not mention switch level redundancy as a requirement. This path is still protected through its dual uplinks to the two core switches and uses open shortest path first to move traffic to the other link if one dies. There is a risk that all cameras connectivity is lost if this distribution switch fails. However, the consequence of this is only a short loss of video stream, which is a lot less severe when compared to losing air traffic control which is why this block has less redundancy. For the general block, this zone has the same logical setup as the CCTV zone, although it does handle a wider range of traffic sources. General includes check in, gate information, tech support, point of sales, management and the various lounges. The traffic from these sources is not critical to safety or life protection. This part of the network also has dual uplinks that connect to both core switches and uses OSPF to handle failure. Even though a failure at this switch would cause the whole terminal to go down, it was imperative to allocate cost to additional switches to other, more vital zones. Additionally, there is no requirement to ensure that this block has redundancy. Because the client wants cost efficient redundancy, I believe that it was important to not double all devices for maximum redundancy as that would have been too costly and decided to only apply redundancy for critical network zones.

WAN Edge Redundancy Justification

This network has a wide area network edge because FR3 states that the network must be able to connect to other major airports across Australia via a WAN. This is key to transmit air traffic information with the other airports. This network needs two routers. If it only had one router, each external path Adelaide has would have to have a router which includes all nine airport links and the internet service provider link. A failure

would disconnect the network from all outside connectivity. FR6 requires the network to operate without the possibility of a single link or path failing that isolates other sites. This would also be a critical link that needs redundancy as mentioned in NFR7. Unlike an access switch failing that only impacts a few devices under it, if the router is a single point of failure the whole network is gone. However, three routers would be too much redundancy as there is no requirement that the network must be able to survive both routers going down at the same time. In this network I decided to go with a full dual-homing setup and prioritise router redundancy over port cost. With this design each airport will get dual links to each router so that if one fails every airport should be able to route through the other remaining routers, satisfying FR6. This also affects hardware sizing. This means that WAN router 1 will need a total of 13 ports, 9 ports for each airport, one ISP port, two core ports and 1 inter-router port. WAN router 2 needs 12. OSPF will also be implemented instead of HSRP. The WAN routers will be in area 0 acting as part of the backbone and the routers will advertise their reachability to the other airports, so the other core devices know two paths to the airports. In the event of a link failing OSPF will find another path for traffic to travel. HSRP would be used if the core couldn't use dynamic routing, and HSRP does not make use of both routers simultaneously leaving one idle. There are two distinct types of routes. Airport connection routes and the route to the internet through the ISP. Neither route can enter OSPF on their own so external routes must be redistributed, forcing the routers to be autonomous system boundary routers advertising type 5 link state advertisements. Adelaide airports design ends at these routers so the OSPF domain does not go further than these devices. OSPF does have limitations, the protocol takes time and uses hello and dead ping timers which can take from 10-40 seconds meaning the network can experience delays while the network tries to detect a neighbour device failure. This latency can be reduced by tuning the timers to custom ping times. There is only a single ISP link because the scenario states that the WAN will be used to re-route internet traffic in the case that local internet is down or slow. This means that the second ISP link is replaced with a mesh design. If the ISP link goes down, then traffic sent to the internet can go through the mesh and reach the internet through the other airports. The throughput of 10G per link is sufficient. With all devices totalling approximately a total of 5Gbps a single router will be able to still handle the traffic.

Melbourne Server Farm Link Justification

There is a direct link dedicated from Melbourne's server farm to Adelaide airport because FR4 requires it. To be a direct link it must have its own link that does not share traffic with the WAN mesh which ensures that it is not competing with other airport traffic and does not depend on routing through other airports. It is distinct from the Adelaide to Melbourne WAN link through the edge router which is the link that meets FR3. There are two links to the server farm. This is because FR4 states that a link must

exist, but NFR7 states that the link to critical services must have dual redundancy. It is my assumption in this scenario that the link to the servers would be classed as a critical link. There is a link on each of the server distribution switches, if they were both on the same switch, that could be a single point of failure. I decided to terminate the server distribution switches instead of the WAN routers. This is because this way the traffic will leave directly from where the servers are without interacting with the core or WAN edge. Because of this, traffic from the servers will not be interrupted if one of the WAN routers fails. In the case that the Melbourne link fails, the redundant link will convey traffic. Each distribution switch has its own static route to the Melbourne server farm, They redistribute the static route into OSPF so other switches learn a path to Melbourne. If a link fails it will take the other learned route which is why these switches are classed as autonomous system boundary routers. If server distribution switch 1 fails, the second goes from the standby to active. If a WAN router or core switch fails, there is no effect on the server link. I decided to route server traffic instead of having it on a stretched VLAN, which ensures that the broadcast domain does not stretch across two whole cities. In this situation it is better to route links for L3 devices rather than use trunking.

Access Layer Sizing Justification

For this network there was no NFR or FR for access layer sizing, but the client did specify that “Copper has distance limitations, so you can't simply group every 'like' device onto a single IDF; how the devices are physically spread across the facility matters.” The intermediate distribution frames are labelled in the physical diagram. From research I conclude that the main terminal is approximately 800 meters long, which is about eight times the maximum length of 100m copper cabling. The level 1 office floor is only about a 20m by 16m area which is within a single copper cables limitations. 100 meters of standard channel does not translate to 100 meters of a floor plan. This is because TIA-568 100m cable is 90m of permanent cable with an additional 10m of patch cord. The 90m is also the real cable length and not the amount of distance the cable would actually cover due to the way the cable would run. This would most likely be through ceiling trays and must be able to go around structures and account for slack loops at each end. This is why for the actual path length I allocated 1.25x of additional length bringing the total length covered by a cable to about 72m and 144m of total area covered per zone. For this design to actually be developed an onsite survey would need to be taken to gain more accurate information. For the terminal's airside, its area is about 800m, when divided by 144m it gives us 5.6 which I rounded up to 6 total zones covering 750-850m. Landside of the terminal is the check in area which is wider rather than long meaning that its network devices can sit within a smaller effective space. Along with the office, the total amount of access switches is 9, which allows for

one access switch per zone. With 8 zones, each zone will have around 15-16 for airside, 38 for landside so they will all stay under the switches 48 port maximum. The only exception is the office which has 50 devices when taking into account the identical level above. There will be two access switches for the office, one for each level. Even if there were less devices the main constraint is cable limitation rather than port count. The reason why there are not dual access switches for each zone is because the damage of losing a zone does not justify the need for 18 switches. Since every zone has cameras that need power over ethernet, each switch needs PoE+ ports which meets FR5.

OSPF Routing Justification

This network will use OSPF routing as its routing protocol. It is not using static routes everywhere because while static routing could theoretically work, there would have to be many primary and floating paths that would all have to be manually set up and maintained. At a network on this scale with all of its built in redundancy, dynamic routing is key to ensure that everything works without lots of manual maintenance. The network also uses single area OSPF instead of multi-area because even though the network is quite large, the total amount of devices being 2 WAN routers, 2 core switches and 6 distribution switches, does not meet the scale of a network where splitting areas is needed. Its network edge encompasses the 10 devices and stops at the WAN routers because the other 9 airports and the ISP link are not managed by Adelaide airport. There are 4 devices in that generate type 5 link state advertisements from external routes which include the two WAN routers and the two server distribution switches from their Melbourne routes.

IP Addressing Scheme

Supernet Table

Site	Supernet	Subnet Mask	Supernet Range	Addresses Required	Total Addresses
Adelaide Airport	10.0.0.0/22	255.255.252.0	10.0.0.0 – 10.0.3.255	652	1,024

Network Table: VLAN Networks

Network	VLAN	Req. / Usable	Network (CIDR)	Subnet Mask	Usable Range	Broadcast
CCTV	10	50 / 62	10.0.0.0/26	255.255.255.192	10.0.0.1 – 10.0.0.62	10.0.0.63
GATES	20	49 / 62	10.0.0.64/26	255.255.255.192	10.0.0.65 – 10.0.0.126	10.0.0.127
ATC-MAIN	30	31 / 62	10.0.0.128/26	255.255.255.192	10.0.0.129 – 10.0.0.190	10.0.0.191

Network	VLAN	Req. / Usable	Network (CIDR)	Subnet Mask	Usable Range	Broadcast
ATC-BKP	31	31 / 62	10.0.0.192/26	255.255.255.192	10.0.0.193 – 10.0.0.254	10.0.0.255
POS	40	30 / 62	10.0.1.0/26	255.255.255.192	10.0.1.1 – 10.0.1.62	10.0.1.63
MGMT	50	20 / 62	10.0.1.64/26	255.255.255.192	10.0.1.65 – 10.0.1.126	10.0.1.127
TECHSUPPORT	60	26 / 30	10.0.1.128/27	255.255.255.224	10.0.1.129 – 10.0.1.158	10.0.1.159
CHECKIN	70	21 / 30	10.0.1.160/27	255.255.255.224	10.0.1.161 – 10.0.1.190	10.0.1.191
SERVERS	80	13 / 30	10.0.1.192/27	255.255.255.224	10.0.1.193 – 10.0.1.222	10.0.1.223
LOUNGES	90	7 / 14	10.0.1.224/28	255.255.255.240	10.0.1.225 – 10.0.1.238	10.0.1.239
CCTV-EXP	11	— / 62	10.0.2.192/26	255.255.255.192	10.0.2.193 – 10.0.2.254	10.0.2.255
OFFICE-EXP	61	— / 30	10.0.3.0/27	255.255.255.224	10.0.3.1 – 10.0.3.30	10.0.3.31
WLAN	100	— / 30	10.0.3.32/27	255.255.255.224	10.0.3.33 – 10.0.3.62	10.0.3.63
Unallocated	10.0.2.140 – 10.0.2.191 (52 addresses: 10.0.2.140/30, 10.0.2.144/28, 10.0.2.160/27) – left by aligning CCTV-EXP to a /26 boundary					
Unallocated – remaining addresses	10.0.3.80 – 10.0.3.255 (176 addresses: 10.0.3.80/28, 10.0.3.96/27, 10.0.3.128/25)					

Network Table: Internal Point-to-Point Links

Network (CIDR)	Side A	IP Address	Side B	IP Address
10.0.1.240/30	CSW-CORE-01	10.0.1.241	CSW-CORE-02	10.0.1.242
10.0.1.244/30	RTR-WAN-01	10.0.1.245	CSW-CORE-01	10.0.1.246
10.0.1.248/30	RTR-WAN-02	10.0.1.249	CSW-CORE-01	10.0.1.250
10.0.1.252/30	RTR-WAN-01	10.0.1.253	CSW-CORE-02	10.0.1.254
10.0.2.0/30	RTR-WAN-02	10.0.2.1	CSW-CORE-02	10.0.2.2
10.0.2.80/30	RTR-WAN-01	10.0.2.81	RTR-WAN-02	10.0.2.82
10.0.2.84/30	CSW-CORE-01	10.0.2.85	DSW-ATC-01	10.0.2.86
10.0.2.88/30	CSW-CORE-01	10.0.2.89	DSW-ATC-02	10.0.2.90
10.0.2.92/30	CSW-CORE-02	10.0.2.93	DSW-ATC-01	10.0.2.94
10.0.2.96/30	CSW-CORE-02	10.0.2.97	DSW-ATC-02	10.0.2.98
10.0.2.100/30	CSW-CORE-01	10.0.2.101	DSW-SRV-01	10.0.2.102

Network (CIDR)	Side A	IP Address	Side B	IP Address
10.0.2.104/30	CSW-CORE-01	10.0.2.105	DSW-SRV-02	10.0.2.106
10.0.2.108/30	CSW-CORE-02	10.0.2.109	DSW-SRV-01	10.0.2.110
10.0.2.112/30	CSW-CORE-02	10.0.2.113	DSW-SRV-02	10.0.2.114
10.0.2.116/30	CSW-CORE-01	10.0.2.117	DSW-CCTV-01	10.0.2.118
10.0.2.120/30	CSW-CORE-02	10.0.2.121	DSW-CCTV-01	10.0.2.122
10.0.2.124/30	CSW-CORE-01	10.0.2.125	DSW-GEN-01	10.0.2.126
10.0.2.128/30	CSW-CORE-02	10.0.2.129	DSW-GEN-01	10.0.2.130
10.0.3.64/28	Reserved pool: 4 x /30 (10.0.3.64/30, 10.0.3.68/30, 10.0.3.72/30, 10.0.3.76/30) for future distribution or WAN links			

Network Table: WAN Point-to-Point Links

Network (CIDR)	Side A	IP Address	Side B	IP Address
10.0.2.4/30	RTR-WAN-01	10.0.2.5	Sydney Airport WAN router	10.0.2.6
10.0.2.8/30	RTR-WAN-02	10.0.2.9	Sydney Airport WAN router	10.0.2.10
10.0.2.12/30	RTR-WAN-01	10.0.2.13	Melbourne Airport WAN router	10.0.2.14
10.0.2.16/30	RTR-WAN-02	10.0.2.17	Melbourne Airport WAN router	10.0.2.18
10.0.2.20/30	RTR-WAN-01	10.0.2.21	Brisbane Airport WAN router	10.0.2.22
10.0.2.24/30	RTR-WAN-02	10.0.2.25	Brisbane Airport WAN router	10.0.2.26
10.0.2.28/30	RTR-WAN-01	10.0.2.29	Perth Airport WAN router	10.0.2.30
10.0.2.32/30	RTR-WAN-02	10.0.2.33	Perth Airport WAN router	10.0.2.34
10.0.2.36/30	RTR-WAN-01	10.0.2.37	Gold Coast Airport WAN router	10.0.2.38
10.0.2.40/30	RTR-WAN-02	10.0.2.41	Gold Coast Airport WAN router	10.0.2.42
10.0.2.44/30	RTR-WAN-01	10.0.2.45	Cairns Airport WAN router	10.0.2.46
10.0.2.48/30	RTR-WAN-02	10.0.2.49	Cairns Airport WAN router	10.0.2.50
10.0.2.52/30	RTR-WAN-01	10.0.2.53	Canberra Airport WAN router	10.0.2.54
10.0.2.56/30	RTR-WAN-02	10.0.2.57	Canberra Airport WAN router	10.0.2.58
10.0.2.60/30	RTR-WAN-01	10.0.2.61	Hobart Airport WAN router	10.0.2.62
10.0.2.64/30	RTR-WAN-02	10.0.2.65	Hobart Airport WAN router	10.0.2.66
10.0.2.68/30	RTR-WAN-01	10.0.2.69	Darwin Airport WAN router	10.0.2.70

Network (CIDR)	Side A	IP Address	Side B	IP Address
10.0.2.72/30	RTR-WAN-02	10.0.2.73	Darwin Airport WAN router	10.0.2.74
10.0.2.76/30	RTR-WAN-01	10.0.2.77	ISP (placeholder)	10.0.2.78
10.0.2.132/30	DSW-SRV-01	10.0.2.133	Melbourne server farm	10.0.2.134
10.0.2.136/30	DSW-SRV-02	10.0.2.137	Melbourne server farm	10.0.2.138

VLAN Table

VLAN	Name	Network	Default Gateway	Gateway Device	Present On
10	CCTV 49 PoE security cameras	10.0.0.0/26	10.0.0.1	DSW-CCTV-01	All 9 access switches (CCTV trunk)
11	CCTV-EXP (reserved) Future security cameras	10.0.2.192/26	10.0.2.193	DSW-CCTV-01	Access switches via CCTV trunk (when deployed)
20	GATES 16 gates × (1 workstation + 2 NUC wallboards)	10.0.0.64/26	10.0.0.65	DSW-GEN-01	ASW-AIRSIDE-01 – 06
30	ATC-MAIN ATC workstations, NIC 1 (30)	10.0.0.128/26	10.0.0.129	DSW-ATC-01	DSW-ATC-01 access ports only (no trunk)
31	ATC-BKP ATC workstations, NIC 2 (30)	10.0.0.192/26	10.0.0.193	DSW-ATC-02	DSW-ATC-02 access ports only (no trunk)
40	POS 29 retail and food & beverage POS devices	10.0.1.0/26	10.0.1.1	DSW-GEN-01	Landside and airside access switches in retail zones
50	MGMT 9 access switches + 10 WAPs	10.0.1.64/26	10.0.1.65	DSW-GEN-01	All 9 access switches (general trunk)
60	TECHSUPPORT 25 technical support workstations	10.0.1.128/27	10.0.1.129	DSW-GEN-01	ASW-OFFICE-01
61	OFFICE-EXP (reserved) Unused upper-floor office space	10.0.3.0/27	10.0.3.1	DSW-GEN-01	ASW-OFFICE-02 (when deployed)
70	CHECKIN 10 workstations + 10 wallboards	10.0.1.160/27	10.0.1.161	DSW-GEN-01	ASW-LANDSIDE-01
80	SERVERS 10 servers (3 x 10G LACP each)	10.0.1.192/27	10.0.1.193	DSW-SRV-01 / DSW-SRV-02 (HSRP)	DSW-SRV-01, DSW-SRV-02 and their inter-switch trunk

VLAN	Name	Network	Default Gateway	Gateway Device	Present On
90	LOUNGES 3 lounges × 2 NUC wallboards	10.0.1.224/28	10.0.1.225	DSW-GEN-01	Airside access switches in the 3 lounge zones
100	WLAN - reserved Wireless clients (no client count given in brief)	10.0.3.32/27	10.0.3.33	DSW-GEN-01	ASW-OFFICE-01, ASW-OFFICE-02 (when deployed)
999	NATIVE Unused native VLAN – no devices	-	-	-	Native VLAN on all 19 trunks

Key IP Addressing Table

Device	Interface	IP Address	Purpose
DSW-CCTV-01	Vlan10	10.0.0.1/26	Default gateway, VLAN 10 CCTV
DSW-GEN-01	Vlan20	10.0.0.65/26	Default gateway, VLAN 20 GATES
DSW-ATC-01	Vlan30	10.0.0.129/26	Default gateway, VLAN 30 ATC-MAIN
DSW-ATC-02	Vlan31	10.0.0.193/26	Default gateway, VLAN 31 ATC-BKP
DSW-GEN-01	Vlan40	10.0.1.1/26	Default gateway, VLAN 40 POS
DSW-GEN-01	Vlan50	10.0.1.65/26	Default gateway, VLAN 50 MGMT
DSW-GEN-01	Vlan60	10.0.1.129/27	Default gateway, VLAN 60 TECHSUPPORT
DSW-GEN-01	Vlan70	10.0.1.161/27	Default gateway, VLAN 70 CHECKIN
HSRP group 80	Vlan80 - virtual	10.0.1.193/27	Default gateway, VLAN 80 SERVERS
DSW-SRV-01	Vlan80	10.0.1.194/27	HSRP active
DSW-SRV-02	Vlan80	10.0.1.195/27	HSRP standby
DSW-GEN-01	Vlan90	10.0.1.225/28	Default gateway, VLAN 90 LOUNGES
DSW-CCTV-01	Vlan11	10.0.2.193/26	Reserved, VLAN 11 CCTV-EXP
DSW-GEN-01	Vlan61	10.0.3.1/27	Reserved, VLAN 61 OFFICE-EXP
DSW-GEN-01	Vlan100	10.0.3.33/27	Reserved, VLAN 100 WLAN
ASW-LANDSIDE-01	Vlan50	10.0.1.66/26	IDF-LANDSIDE-01, landside check-in
ASW-AIRSIDE-01	Vlan50	10.0.1.67/26	IDF-AIRSIDE-01, airside zone 1
ASW-AIRSIDE-02	Vlan50	10.0.1.68/26	IDF-AIRSIDE-02, airside zone 2
ASW-AIRSIDE-03	Vlan50	10.0.1.69/26	IDF-AIRSIDE-03, airside zone 3

Device	Interface	IP Address	Purpose
ASW-AIRSIDE-04	Vlan50	10.0.1.70/26	IDF-AIRSIDE-04, airside zone 4
ASW-AIRSIDE-05	Vlan50	10.0.1.71/26	IDF-AIRSIDE-05, airside zone 5
ASW-AIRSIDE-06	Vlan50	10.0.1.72/26	IDF-AIRSIDE-06, airside zone 6
ASW-OFFICE-01	Vlan50	10.0.1.73/26	IDF-OFFICE-01, Level 01 Offices
ASW-OFFICE-02	Vlan50	10.0.1.74/26	IDF-OFFICE-02, upper office floor
WAP-OFFICE-01 – 05	VLAN 50	10.0.1.80 – 10.0.1.84	Level 01 Offices, via ASW-OFFICE-01
WAP-OFFICE-06 – 10	VLAN 50	10.0.1.85 – 10.0.1.89	Upper office floor, via ASW-OFFICE-02
Reserved	VLAN 50	10.0.1.75 – 10.0.1.79	Future access switches
Reserved	VLAN 50	10.0.1.90 – 10.0.1.99	Future access points
RTR-WAN-01	To CSW-CORE-01	10.0.1.245/30	Management (alternate 10.0.1.253)
RTR-WAN-02	To CSW-CORE-01	10.0.1.249/30	Management (alternate 10.0.2.1)
CSW-CORE-01	Core ISL	10.0.1.241/30	Management (alternate 10.0.2.125)
CSW-CORE-02	Core ISL	10.0.1.242/30	Management (alternate 10.0.2.129)
DSW-ATC-01	To CSW-CORE-01	10.0.2.86/30	Management (alternate 10.0.2.94)
DSW-ATC-02	To CSW-CORE-01	10.0.2.90/30	Management (alternate 10.0.2.98)
DSW-SRV-01	To CSW-CORE-01	10.0.2.102/30	Management (alternate 10.0.2.110)
DSW-SRV-02	To CSW-CORE-01	10.0.2.106/30	Management (alternate 10.0.2.114)
DSW-CCTV-01	To CSW-CORE-01	10.0.2.118/30	Management (alternate 10.0.2.122)
DSW-GEN-01	Vlan50	10.0.1.65/26	Management (alternate 10.0.2.126)

Variable length subnet mask and IP Addressing

Justification

This network requires VLSM because the client explicitly requires it. An IP addressing scheme like 10.site.VLAN.0/24 would be too broad and generic. A VLSM is also required in this network to be able to fit the 49 serviceable sub-networks which include the 10 VLANs and 39 point to point links with 2 to 62 usable hosts. Therefore a /26 that is fixed

for all of them would need 3136 addresses ($49 * 64$) which would be /20. VLSM fits the same networks in 652 which is /22. This is done by ordering the networks from largest to smallest and giving each network the smallest block that fits the network (with the exclusion of MGMT and POS). To ensure that there are no gaps or overlapping parts in the layout you order by largest first which keeps all blocks aligned to a boundary matching its own size. To size the networks for each end user it considers all that networks devices and adds one address for the gateway, then converts to the smallest usable block. It is slightly different with the servers however. Because they use HSRP, their block required 10 server addresses, a shared virtual gateway and two additional real interface addresses, one for each server distribution switch. This is because HSRP requires the switches to have their own address and the shared virtual gateway. Air traffic control also needs its gateway along with its 30 devices meaning this needs a /26. The lounges have 7 addresses in total including the gateway which requires it to have a /28. All point to point links will use /30 which is just enough room for 2 usable addresses per end. With this the network should be able to accommodate all hosts including 20 for check in, 25 for tech support, 10 servers, 30 ATC workstations, 29 points of sale and 6 lounges as per the terminal map. It was also assumed that there would be 16 gates and 49 CCTV cameras. For this network there will be a supernet of 10.0.0.0/22. The end user networks will be allocated first from 10.0.0.0 to 10.0.1.239. The point to point addresses will follow from 10.0.1.240 to 10.0.2.139, and the remaining addresses can be reserved for further headroom expansion. /22 is used because the networks smallest total size is 652 which is greater than 512 for a /23. This is a contiguous block meaning that the rest of the WAN uses a single route of 10.0.0.0/22 to reach the Adelaide network. In this network the gateway will be the first usable address for each network. For the servers this means that its first address will be used as the HSRP virtual IP. In this design the access uplinks are not included in the table because they are 802.1Q trunks and are not routed. The VLANS for this network are 10, 11, 20, 30, 31, 40, 50, 60, 61, 70, 80, 90, 100 and native VLAN 999. There is a VLAN for each end user network and each gateway for the distribution switches. It is assumed that 10.0.0.0/8 is private and other airports are assumed to have their own blocks outside of Adelaide's design.

Network Hardware and Devices

Network Hardware Choices

Device	Model	Qty	Ports Used / Available	Capacity and Power	Lifecycle (Sep 2026)
RTR-WAN-01, RTR-WAN-02	C8650-G2 Cisco 8650-G2 Secure Router (Cisco, 2026)	2	10G SFP+: 11 of 20; 100G QSFP28: 2 of 6 (RTR-WAN-02 uses 10 × 10G)	Up to 540 Gbps forwarding; 1100 W AC PSUs, N+2	Current (released June 2026); Cisco's replacement for the end-of-life C8500-20X6C
CSW-CORE-01, CSW-CORE-02	C9500X-28C8D Cisco Catalyst 9500X (Cisco, 2026)	2	400G QSFP-DD: 4 of 8; 100G QSFP28: 5 of 28	Non-blocking 400G ports; 1+1 AC PSUs	Current; no end-of-sale notice
DSW-ATC-01, DSW-ATC-02	C9500X-60L4D Cisco Catalyst 9500X (Cisco, 2026)	2	10G SFP56: 30 of 60; 400G QSFP-DD: 2 of 4	Non-blocking; one 400G uplink carries all 300 Gbps of ATC traffic (NFR2); 1+1 AC PSUs	Current; no end-of-sale notice
DSW-SRV-01, DSW-SRV-02	C9500X-60L4D Cisco Catalyst 9500X (Cisco, 2026)	2	10G SFP56: 16 of 60; 400G QSFP-DD: 4 of 4	2 × 400G uplinks plus 2 × 400G inter-switch trunk; 1+1 AC PSUs	Current; no end-of-sale notice
DSW-CCTV-01, DSW-GEN-01	C9550-24L4CD Cisco C9550 Smart Switch (Cisco, 2026)	2	10G SFP56: 9 of 24; 100G QSFP-DD: 2 of 4	Up to 2.4 Tbps; 1+1 AC PSUs	Current (released June 2026); successor to the Catalyst 9500H
ASW-LANDSIDE-01, ASW-AIRSIDE-01 – 06, ASW-OFFICE-01, ASW-OFFICE-02	C9350-48P Cisco C9350 Smart Switch (Cisco, 2026)	9	1G PoE+: up to 40 of 48; 10G uplinks (C9350-NM-8Y): 2 of 8	30 W PoE+ per port, 1,440 W PoE budget (FR5)	Current; successor to the Catalyst 9300 1G PoE models

Supplementary Equipment

Part Number	Description	Qty	Used For
C9350-NM-8Y	8 × 1/10/25G SFP28 network module (Cisco, 2026)	9	One per access switch for its two 10G uplinks (the switch ships without an uplink module)
QDD-400G-FR4-S	400G QSFP-DD, 2 km SMF, duplex LC (Cisco, 2026)	16	Core QSFP-DD ↔ DSW-ATC and DSW-SRV QSFP-DD: 8 links, both ends
SFP-10G-LR-S	10G SFP+, 10 km SMF, duplex LC (Cisco, 2026)	57	Access uplinks, NM-8Y ↔ DSW-GEN/DSW-CCTV SFP56: 18 links, both ends (36). Adelaide end of the 18 airport links, the ISP link and the 2 Melbourne links (21)
SFP-10G-SR-S	10G SFP+, up to 400 m OM4, duplex LC (Cisco, 2026)	60	Switch end of the 60 ATC workstation links (workstation NIC optics are supplied with the workstations)

Part Number	Description	Qty	Used For
QDD-400-CU2M	400G QSFP-DD passive DAC, 2 m (Cisco, 2026)	2	DSW-SRV-01 ↔ DSW-SRV-02 inter-switch trunk (2-link EtherChannel)
QSFP-100G-CU3M	100G QSFP28 passive DAC, 3 m (Cisco, 2026)	9	Core ↔ WAN routers (4), core ISL (1), core ↔ DSW-CCTV-01 and DSW-GEN-01 (4)
SFP-H10GB-CU3M	10G SFP+ passive DAC, 3 m (Cisco, 2026)	31	Server connections (30: 3 per server, LACP) and the RTR-WAN inter-router link (1)

Network Hardware Justification

NFR1 requires the network to continue to use Cisco. Since existing hardware is at end of support, devices chosen will not be nearing end of sale, and will be currently sold and supported. With the C8500-20X6C, its last order date is 31/07/2027 and will be replacing the current routers which is Cisco's recommended migration. The C9350 and C9550 will also be used as they were made available this year. NFR2 requires simultaneous 10G speeds from air traffic control to the server farm for each workstation. This means that the network needs to handle 300Gbps on this link. All workstations normally use 1 NIC, so ATC's distribution switch alone needs to transmit the whole 300Gbps. Only two uplinks of 100G would not be sufficient. This is why 400G uplinks is the number required. Each ATC switch will have a 400G connection to each core so a single link can satisfy NFR2 when something goes down. The model chosen for this is the C9500X-60L4D. This device provides 60 ports of 10/25/50G and four 400G QSFP-DD ports. NFR3 states that each of the 9 major airports needs two 10G link to each other. The 8650-G2 has 20 x 10G SFP+ and 6 x 40/100G QSFP28 ports with a forward max of 540 Gbps. This is needed for wan router 1, using 11 10G ports and 2 x 100G ports. NFR4 requires each device to have 20Mbps for internet access. For about 250 network devices that requires a total of 5Gbps which can be handled by a single 10G ISP link. NFR5 states that each camera streams 100Mbps to the server farm. At 49 cameras, their throughput is 4.9Gbps and is easily covered by the two 100G links on CCTV's distribution switch. Because 100G is way more than the 4.9Gbps required both ends can use QSFP ports, so no breakout cables are needed and there is ample room for expansion. FR5 requires the cameras to be powered over ethernet, thus using the C9350-48P for its 48 ports of 1G and 30 watt PoE+ should be able to cover the power requirements of the cameras and access points. Because the C9350 does not have an uplink module each switch is also paired with a C9350-NM-8Y for its two uplinks of 10G. It is assumed that ATC's NIC's are using fibre because copper 10G SFP uses 2.5W and can only reach 30 meters at 10G.

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